FUNCTION ARITHMETIC

Combining outputs vs chaining functions

Arithmetic combines outputs at the same input; composition feeds one output into the next function.

Arithmetic combination

RULE

$(f+g)(x) = f(x) + g(x)$, and likewise for $-$, $\times$, $\div$

ACTS ON

On the outputs: the input x passes through both functions, then the outputs combine

EXAMPLE

$f(x) = x^2$, $g(x) = 3x$: $(f+g)(3) = 9 + 9 = 18$

RESULT

A single new function with its own formula, $(f+g)(x) = x^2 + 3x$

Composition

RULE

$(f \circ g)(x) = f(g(x))$

ACTS ON

Chains the functions: the output of g becomes the input of f

EXAMPLE

$f(g(3)) = f(9) = 81$

RESULT

A new function built by chaining, covered on the **composition** page

≠ NOT THE SAME

Combining outputs is not the same as chaining functions.

All four arithmetic operations follow one rule, operate on the outputs and keep the input; composition changes the input of the outer function.